

OPEN RESEARCH PROTOTYPE / CONCEPT NOTE

Mexico Scientific Intelligence

Evidence infrastructure for transparent regional capability, science-to-market, and investment analysis.

HADOX RESEARCH LABS

Version 1.0 - July 2026

Apache-2.0 open core

Purpose

Mexico Scientific Intelligence is an evidence infrastructure concept for turning fragmented regional, economic, scientific, and institutional information into traceable capability profiles and reviewable opportunity hypotheses. It is designed for regional development teams, research institutions, innovation intermediaries, investors, and public-interest programs that need to understand where scientific capacity can become a product, pilot, partnership, or investable program.

This concept note is grounded in the contracts and reference source published in this repository. It does not claim that every data layer or commercial conclusion is complete.

The problem

Regional opportunity studies often fail in one of two ways. They remain descriptive reports that cannot be updated, or they compress uncertain evidence into rankings whose assumptions are difficult to inspect. Useful decision infrastructure must preserve provenance, distinguish official evidence from derived interpretation, and connect analysis to execution.

The proposed system

The reference implementation organizes a regional package into a source manifest, core market tables, detailed evidence tables, analytical modules, reports, and workflow artifacts. Nine module contracts connect the evidence to decisions:

- 1 Investment Dashboard
- 2 Market Intelligence
- 3 Science Capacity
- 4 Product Lines
- 5 Roadmap
- 6 Priority Execution
- 7 Grounded Packages
- 8 90-Day Board
- 9 Pilot Clients

The intent is not to automate judgment away. It is to make each material claim inspectable and to expose missing evidence before a recommendation is treated as diligence.

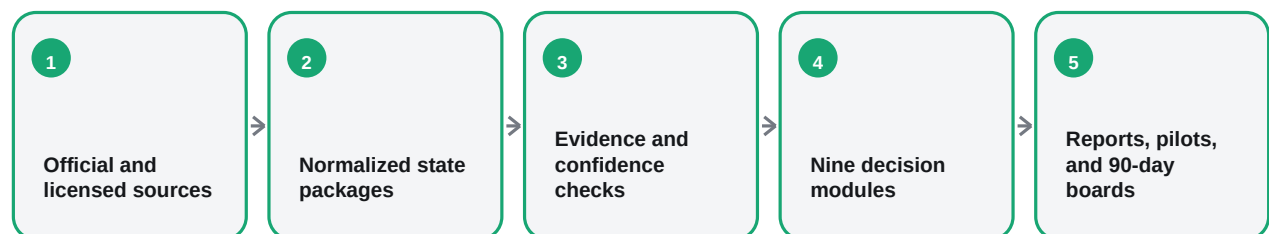
What exists in the open core

- A generalized country-platform data contract and region package layout
- A module plugin contract and registry-driven analytical model
- Reusable source-ledger, derived-KPI, and data-quality templates
- Reference configuration for Morelos, Queretaro, and Tlaxcala
- Service references for Postgres and file adapters, report registration, audit events, and a reusable RBAC vocabulary
- Rules for separating public outputs from private enrichment and operational records

Architecture

The system follows a staged evidence path:

Cross-cutting references provide a role and permission vocabulary, audit events, report registration, and persistent storage. Authentication is deliberately outside this release: deployments should integrate a maintained identity provider or a separately reviewed implementation. Country-specific evidence stays separate from reusable framework contracts so a new implementation must rebuild its claims rather than copy another country's conclusions.



Evidence and governance

Every package should identify source, date, license, generation time, and known limitations. Derived indicators should publish their method, input completeness, confidence boundaries, and review status. Personal contact data, private entity crosswalks, and unapproved named outreach lists belong outside the public repository.

The original prototype demonstrated the shape of the system, but some historic report bundles contained operational and personal fields. They are deliberately excluded from this open export. Future examples should use synthetic or explicitly redistributable evidence.

Product opportunity

The architecture can support repeatable state capability briefs, science-to-market opportunity screens, regional investment narratives, consortium design, pilot pipelines, and governed analytical services. A commercial implementation would combine the open contracts with maintained source connectors, review operations, client-specific evidence, and service-level commitments.

Current maturity

This release is an open architectural and methodological baseline. It is not a finished diligence product or a turnkey deployment. The former interface depended on a separately licensed administrative template and is not redistributed here; its custom authentication and administration code is also excluded because it is not an endorsed security reference. Data coverage, entity resolution, and commercial validation still require source-specific work and human review.

Maturity statement: open methodological baseline; human review and source-specific validation remain mandatory.

Open contribution roadmap

- Create a template-independent accessible web interface
- Add official-source adapters and automated provenance checks
- Publish synthetic region packages and validation tests
- Improve confidence, bias, and uncertainty representation
- Add privacy-safe entity resolution and approval workflows
- Define interoperable APIs for external compute and analytical agents
- Develop reproducible evaluation benchmarks for regional recommendations

Repository evidence index

- `docs/platform/country-platform-data-contract.md`
- `docs/platform/module-plugin-contract.md`
- `docs/platform/country-data-acquisition-and-enrichment-runbook.md`
- `docs/platform/templates/`
- `reference/framework/config/intelligenceFramework.js`
- `reference/modules/mx-states/reportConfig.js`
- `reference/server-services/`
- `tools/build_concept_note.py`

Open-source terms

The clean Hadox-authored export is available under Apache-2.0. Contributions are welcome through GitHub issues and pull requests. Third-party components and external source data remain subject to their own terms.

Repository: <https://github.com/ekaropolus/hadox-scientific-intelligence-mx>

Project page: <https://hadox.org/work/mexico-scientific-intelligence>